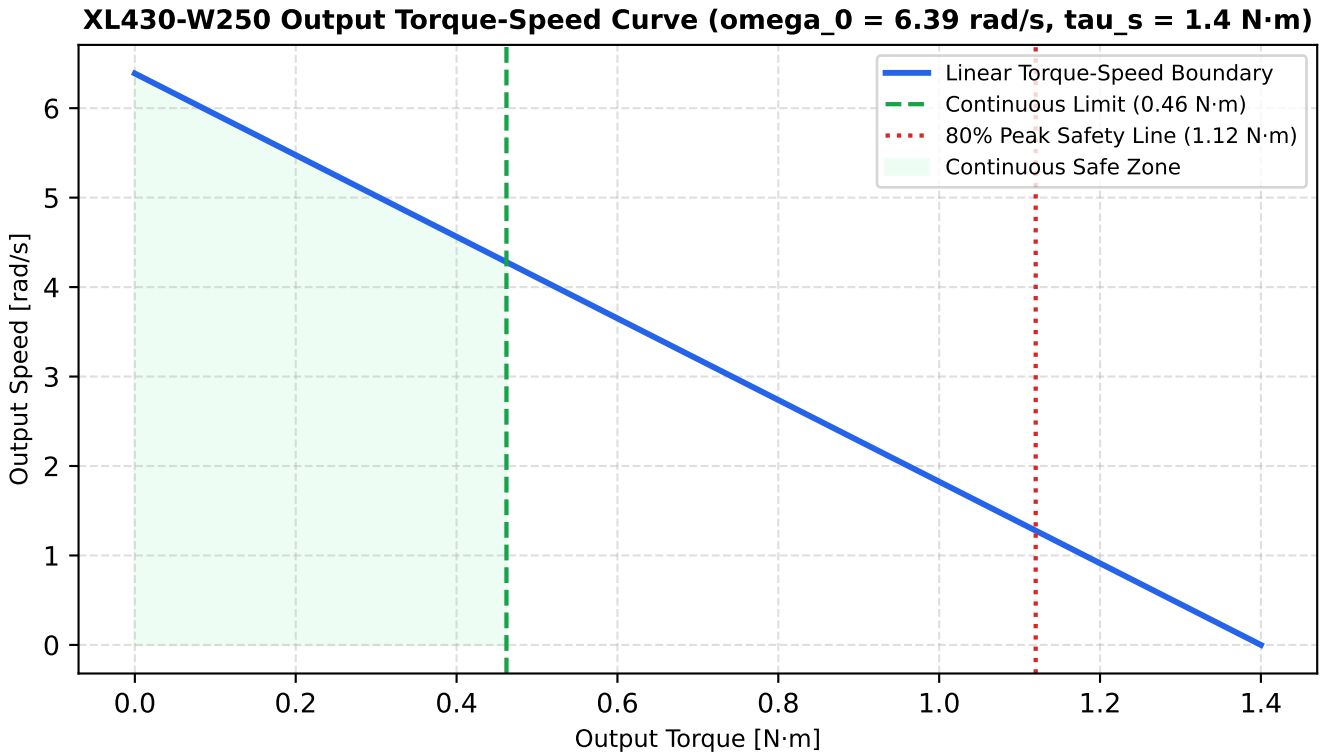
Part Number: XL430-W250

1. Electrical & Mechanical Specifications (Standard Reference)

Parameter	Symbol	Specification Value	Test Condition / Notes
Part Number	ID	XL430-W250	DYNAMIXEL-X
Manufacturer	Vendor	ROBOTIS	Cored DC Motor
Rated Input Voltage	V_rated	12.0 V	Nominal DC Bus
Stall / Peak Torque	tau_stall	1.4 N.m (at 12.0V, 1.3A)	At 12.0 V
Continuous Rated Torque	tau_cont	Not specified (use thermal der	25 degC ambient, continuous duty
No-Load Speed	omega_0	61 rev/min (at 12.0V)	Unloaded output shaft
Actuator Mass (Weight)	m_act	57.2 g	Includes housing & internal gearhead
Internal Reduction	i_int	258.5 : 1	Integrated gear stage
Envelope Dimensions	W x H x D	28.5 x 46.5 x 34.0 mm	Outer housing envelope



XL430-W250 — Mechanical Drawing & Operating Guidelines

2. Mechanical Mounting & Frame Integration
   - Housing Envelope: 28.5 mm x 46.5 mm x 34.0 mm
   - Total Verified Mass: 57.2 g (0.0572 kg)
   - Mounting Pattern: M2.5 / M3 ISO radial & horn flange threads

3. Thermal & Duty-Cycle Notes
   - Continuous operation must remain at or below the continuous rated torque.
   - If continuous torque is not explicitly tabulated on Page 1, apply a 0.33x stall torque
   - Peak torque operation (>80% of available speed-dependent torque) risks step loss or the

4. Citation Verification Hash
   - Part ID: XL430-W250 | Source Document: xl430-w250.pdf | Spec Table: Page 1